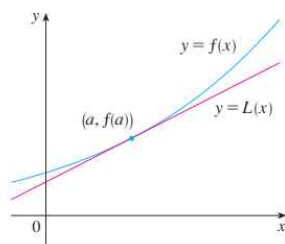


## §8 Linear Approximations and Differentials

- \* An equation of the tangent line to the curve  $y = f(x)$  at  $(a, f(a))$  is  
$$y = f(a) + f'(a)(x - a).$$



For a given number  $x$  near  $a$ , the approximation

$$f(x) \approx f(a) + f'(a)(x - a)$$

is called the **linear approximation** or **tangent line approximation** of  $f$  at  $a$ , and the linear function

$$L(x) = f(a) + f'(a)(x - a)$$

is called the **linearization** of  $f$  at  $a$ .

**Exam 1** Find the linearization of the function  $f(x) = \sqrt{x+3}$  at  $a=1$  and use it to approximate the numbers  $\sqrt{3.98}$  and  $\sqrt{4.05}$ . Are these approximations overestimates or underestimates?

$$\text{(sol)} \quad f(x) = \sqrt{x+3} = (x+3)^{\frac{1}{2}} \Rightarrow f(1) = 2$$

$$\Rightarrow f'(x) = \frac{1}{2}(x+3)^{-\frac{1}{2}} = \frac{1}{2\sqrt{x+3}} \Rightarrow f'(1) = \frac{1}{4}$$

Then the linearization of  $f$  at  $a=1$  is

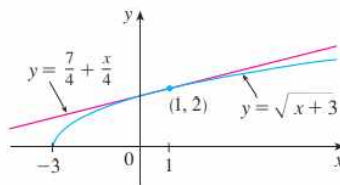
$$\begin{aligned} L(x) &= f(1) + f'(1)(x-1) \\ &= 2 + \frac{1}{4}(x-1) \\ &= \frac{1}{4}x + \frac{7}{4} \end{aligned}$$

and the corresponding linear approximation of  $f$  at  $a=1$  is

$$\sqrt{x+3} \approx \frac{1}{4}x + \frac{7}{4}.$$

$$\Rightarrow \textcircled{1} \quad x = 0.98: \sqrt{3.98} \approx \frac{1}{4} \times 0.98 + \frac{7}{4} = 1.995 \quad (1.9949937)$$

$$\textcircled{2} \quad x = 1.05: \sqrt{4.05} \approx \frac{1}{4} \times 1.05 + \frac{7}{4} = 2.0125 \quad (2.0124611)$$



These approximations are overestimates because the tangent line lies above the curve.

**Exam 2** For what values of  $x$  is the linear approximation

$$\sqrt{x+3} \approx \frac{1}{4}x + \frac{7}{4}$$

accurate to within 0.5? What about accuracy to within 0.1?

(sol) (1) Accuracy to within 0.5.

$$\left| \sqrt{x+3} - \left( \frac{7}{4} + \frac{x}{4} \right) \right| < 0.5$$

$$\Rightarrow -0.5 < \frac{7}{4} + \frac{x}{4} - \sqrt{x+3} < 0.5$$

$$\Rightarrow \sqrt{x+3} - 0.5 < \frac{7}{4} + \frac{x}{4} < \sqrt{x+3} + 0.5$$

$$\left( \begin{array}{l} \frac{7}{4} + \frac{x}{4} = \sqrt{x+3} + \frac{1}{2} \\ \Rightarrow x+5 = 4\sqrt{x+3} \\ \Rightarrow x^2 - 6x - 23 = 0 \\ \Rightarrow x = 3 \pm 4\sqrt{2} \end{array} \right)$$

$$\Rightarrow 3 - 4\sqrt{2} < x < 3 + 4\sqrt{2}$$

$$\Rightarrow -2.656 < x < 8.656$$

$$\Rightarrow -2.65 < x < 8.65$$

(2) Accuracy to within 0.1.

$$\left| \sqrt{x+3} - \left( \frac{7}{4} + \frac{x}{4} \right) \right| < 0.1$$

$$\Rightarrow \sqrt{x+3} - 0.1 < \frac{7}{4} + \frac{x}{4} < \sqrt{x+3} + 0.1$$

$$\Rightarrow -1.129 < x < 3.928$$

$$\Rightarrow -1.12 < x < 3.92$$

### \* Differentials

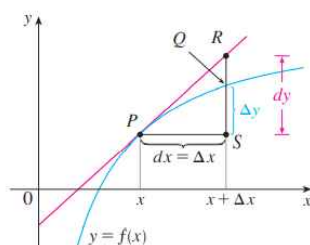
Suppose that  $y = f(x)$  is a differentiable function.

The **differential**  $dx$  is an independent variable and the **differential**  $dy$  is defined in terms of  $dx$  by the equation

$$dy = f'(x) dx .$$

#### • The geometric meaning of differentials

Let  $P(x, f(x))$  and  $Q(x + \Delta x, f(x + \Delta x))$  be two points on the graph of  $y = f(x)$  and let  $dx = \Delta x$



The corresponding change in  $y$  is  $\Delta y = f(x + \Delta x) - f(x)$ .

The slope of the tangent line  $PR$  is the derivative  $f'(x)$ .

Then  $\frac{SR}{PS} = f'(x)$ .

$$\Rightarrow SR = f'(x) PS = f'(x) \Delta x = f'(x) dx = dy .$$

$\Rightarrow$  The approximation  $\Delta y \approx dy$  becomes better as  $\Delta x$  becomes smaller.

- In the linear approximation

$$f(x) \approx f(a) + f'(a)(x-a).$$

Putting  $dx = x - a$ , we have

$$f(a+dx) \approx f(a) + f'(a)dx.$$

For an arbitrary  $x \in \mathbb{R}$ ,

$$\begin{aligned} f(x+dx) &\approx f(x) + f'(x)dx \\ &= f(x) + dy \end{aligned}$$

- For instance, for  $f(x) = \sqrt{x+3}$  in Example 1,  $f'(x) = \frac{1}{2\sqrt{x+3}}$

and so

$$\sqrt{x+3+dx} \approx \sqrt{x+3} + \frac{dx}{2\sqrt{x+3}}$$

(a) Taking  $x=1$  and  $dx=0.05$ , we have

$$\begin{aligned} \sqrt{4.05} &\approx \sqrt{4} + \frac{0.05}{2\sqrt{4}} \\ &= 2 + \frac{0.05}{4} \\ &= 2.0125 \end{aligned}$$

(b) Taking  $x=6$  and  $dx=0.01$ , we have

$$\begin{aligned} \sqrt{9.01} &\approx \sqrt{9} + \frac{0.01}{2\sqrt{9}} \\ &= 3 + \frac{0.01}{6} \\ &= 3.0016 \dots \end{aligned}$$

(cf) In Example 1, we have the linear approximation  $\sqrt{x+3} \approx \frac{1}{4}x + \frac{7}{4}$ .

Taking  $x=6.01$ , we have

$$\begin{aligned} \sqrt{9.01} &\approx \frac{7}{4} + \frac{6.01}{4} \\ &\approx 3.2526, \end{aligned}$$

which is not good approximation.

The real value of  $\sqrt{9.01}$  is  $3.0016662039 \dots$ .

**Exam 3** The radius of a sphere was measured and found to be 21 cm with a possible error in measurement of at most 0.05 cm. What is the maximum error in using this value of the radius to compute the volume of the sphere?

(sol) If the radius of the sphere is  $r$ , then its volume is  $V = \frac{4}{3}\pi r^3$ . If the error in the measured value of  $r$  is denoted by  $dr = \Delta r$ , then the corresponding error in the calculated value of  $V$  is  $\Delta V$ , which can be approximated by the differential

$$dV = 4\pi r^2 dr.$$

When  $r = 21$  and  $dr = 0.05$ , this becomes

$$dV = 4\pi(21)^2 0.05 \approx 277.$$

The maximum error in the calculated volume is about  $277 \text{ cm}^3$ .

(Note) In Example 3,

$$\frac{dV}{V} = \frac{4\pi r^2 dr}{\frac{4}{3}\pi r^3} = \frac{3dr}{r} = \frac{3 \times 0.05}{21} = 0.007$$

is called the **relative error** in the volume and

$$\frac{dV}{V} \times 100 = 0.7 \%$$

is called the **percentage error** in the volume.

**Exercise 24** Use differentials to estimate the amount of paint needed to apply a coat of paint 0.05 cm thick to a hemispherical dome with diameter 50 m.

(sol) The correct total amount of paint needed is

$$\frac{1}{2} \left\{ \frac{4}{3} \pi (25.0005)^3 - \frac{4}{3} \pi (25)^3 \right\} \approx 1.96353 \text{ (m}^3\text{)}.$$

The volume of the hemispherical dome with radius  $r$  is  $V = \frac{2}{3} \pi r^3$ .

$\Rightarrow$  The differential  $dV$  of the volume  $V$  is  $dV = 2\pi r^2 dr$ .

$\therefore$  The correct total amount of paint needed is approximately about

$$\begin{aligned} dV|_{r=25, dr=0.0005} &= 2\pi \cdot 25^2 \cdot 0.0005 \\ &= \frac{5}{8} \pi \text{ (m}^3\text{)} \\ &\approx 1.96350 \text{ (m}^3\text{)} \\ &= 2 \text{ (m}^3\text{)}. \end{aligned}$$